Chapter 4: Mechanical Oscillations

I-Periodic motion

It is a movement that repeats itself identically during a constant time interval or a constant duration called period of symbol T.

The unit of the period in the SI system is the second (s).

Frequency is the number of cycles or periods per unit of time. Its symbol is f and it is expressed in hertz (Hz).

Period: $T = \frac{duration\ for\ n\ oscillations}{number\ of\ the\ oscillations} = \frac{\Delta t}{n}$

Frequency: $f = \frac{number\ of\ the\ oscillations}{duration\ for\ theses\ oscillations} = \frac{n}{\Delta t}$

I.e. that the relation between f and T is: $T = \frac{1}{f}$ $\quad or \quad$ $f = \frac{1}{T}$

II-Types or modes of mechanical oscillations

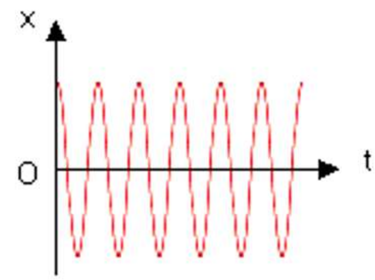
A- Free mechanical oscillations

We provide the oscillator with some energy once and then let it go on its own.

i- Undamped oscillations – Periodic regime

Friction do not exist so there is conservation of mechanical energy:

- ME = constant, thus $\frac{dME}{dt} = 0 \quad \forall\, x \ and\ t$
- The amplitude X_m of the oscillations remains constant.
- The regime of the oscillations is said periodic of proper period T_0.

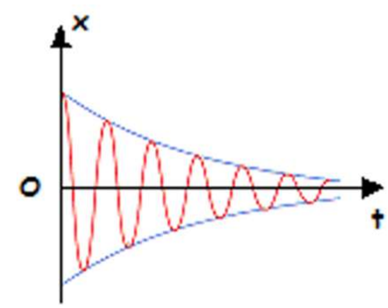


No Friction; the phenomenon is periodic

ii- Low damping oscillations – Pseudo-periodic regime

Friction exists but it is weak (less important):

- Mechanical energy cannot be conserved. It decreases with x & t
- $\Delta ME = W_{\vec{f}}$ thus $\frac{dME}{dt} = (W_{\vec{f}})' = Power\ of\ \vec{f}$
- The amplitude X_m decreases with time t.

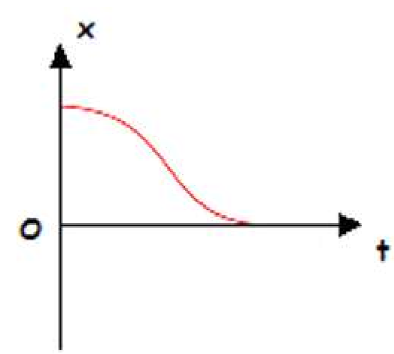


Low friction; pseudo-periodic regime of pseudo-period (or false period) T slightly larger than T_0

iii- Very damped oscillations – Aperiodic regime

Frictions are very important :

- ME decreases quickly. ME is not conserved.
- The amplitude X_m decreases rapidly with the time t.
- The oscillation regime is called aperiodic.



Important friction. The regime is aperiodic

B- Maintained mechanical oscillations

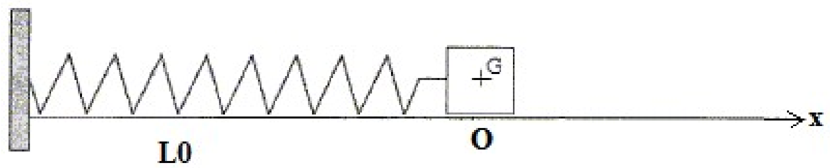
They are mechanical oscillations with little friction. They are maintained by supplying the oscillator with energy to compensate the mechanical energy loss:

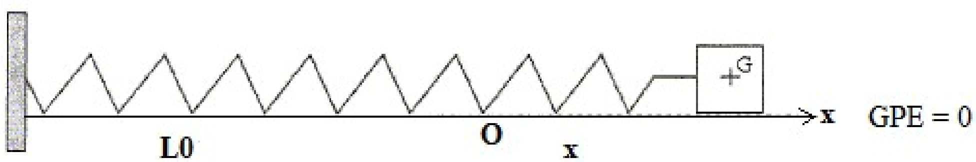
Energy supplied to the oscillator during a period is $W = |\Delta E_m|$
The mechanical power supplied to the oscillator is $P_{average} = \frac{W}{\Delta t} = \frac{W}{T}$

C- Forced oscillations

They are subject of chapter 5 (reduced for SV class)

III- <u>Analytical study of the motion of an undamped elastic horizontal pendulum</u>





A-Energetic method

At a certain moment t, the expression of the ME of the system (spring, block, Earth) is given by:
$ME = KE + GPE + EPE = \frac{1}{2}mv^2 + 0 + \frac{1}{2}K\Delta L^2$ where: $\Delta L = L_{(t)} - L_0 = L_0 + x - L_0 = x$
So $ME = \frac{1}{2}mv^2 + \frac{1}{2}Kx^2$

No friction, then ME = constant:

Therefore $\frac{dME}{dt} = 0$ (the derivative of the ME with respect to t equals 0) $\Rightarrow (\frac{1}{2}mv^2 + \frac{1}{2}Kx^2)' = 0$

$\Rightarrow \frac{1}{2}m2vv' + \frac{1}{2}K2xx' = 0$ or $mva + Kxv = 0 \Rightarrow a = -\frac{K}{m}x$

Since x is variable, K and m are two constants, it follows that the acceleration a varies with t and x, as result the motion varied rectilinear.

In addition, since the motion is rectilinear, then $v = x'$ and $a = v' = x''$
$a = -\frac{K}{m}x$ becomes then: $x'' = -\frac{K}{m}x$ or $x'' + \frac{K}{m}x = 0$ this is a differential equation of second order in x without a second member. It has the form: $x'' + \omega_0^2 x = 0$ with $\omega_0^2 = \frac{K}{m}$

B-Dynamic method:

System: block of center of inertia G

External forces: $m\vec{g}$, $\vec{N}$ and $\vec{T}$

Let's apply Newton's second law on G: $\sum \vec{F}_{ext} = m.\vec{a}$ or $m\vec{g} + \vec{N} + \vec{T} = m\vec{a}$ then $\vec{T} = m\vec{a}$
so $\vec{T}$ and $\vec{a}$ are collinear in the same direction ($m > 0$), so we can write algebraically $T = ma \Rightarrow -K.x = ma$
$\Rightarrow a = -\frac{K}{m}x \Rightarrow x'' + \frac{K}{m}x = 0$

It is a differential equation of second order in x without second member. It has the form: $x'' + \omega_0^2 x = 0$
with $\omega_0^2 = \frac{K}{m}$

Note: The algebraic value of the spring tension (or restoring force) is given by Hooke's law: $T = -K.x$
T and x are always of different signs.

IV-Solution of the differential equation:

The solution of the differential equation is of the form: $x(t) = X_m \cdot \sin(\omega_0 t + \varphi)$

x : is called elongation of the mobile at time t, in m.

X_m : amplitude of the oscillations or maximum elongation, in m ($X_m > 0$).

ω_0 : proper pulsation or natural angular frequency, in rad/s.

T_0: proper period of oscillations, in s; $T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\sqrt{\frac{K}{m}}} = 2\pi\sqrt{\frac{m}{K}}$

f_0: proper frequency of oscillations, in Hz; $f_0 = \frac{1}{T_0} = \frac{\omega_0}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{K}{m}}$

$\omega_0 t + \varphi$: phase at the instant t, in rad.

φ : phase at the origin of the time, in rad. It depends on the initial conditions x_0 and v_0

$-1 \leq \sin(w_0 t + \varphi) \leq 1$; by multiplying by $X_m \Rightarrow -X_m \leq X_m \sin(w_0 t + \varphi) \leq X_m \Rightarrow -X_m \leq x(t) \leq X_m$

V-Relationship between x and v independent of t

1st method: Elimination of t

$x(t) = X_m \cdot \sin(\omega_0 t + \varphi)$; $v = x' = X_m \cdot \omega_0 \cos(\omega_0 t + \varphi)$

It follows that: $\sin(\omega_0 t + \varphi) = \frac{x}{X_m}$ and $\cos(\omega_0 t + \varphi) = \frac{v}{X_m \cdot \omega_0}$

Since $sin^2(\omega_0 t + \varphi) + cos^2(\omega_0 t + \varphi) = 1$, then $\frac{x^2}{X_m^2} + \frac{v^2}{X_m^2 \omega_0^2} = 1$; finally, $x^2\omega_0^2 + v^2 = X_m^2\omega_0^2$

2nd method: Conservation of *ME*

$ME(x, v) = ME(x = \pm X_m, v = 0)$

$\frac{1}{2}mv^2 + \frac{1}{2}Kx^2 = \frac{1}{2}KX_m^2$, with $k = m.\omega_0^2$, then $mv^2 + m\omega_0^2 x^2 = mX_m^2\omega_0^2$ or $v^2 + x^2\omega_0^2 = X_m^2\omega_0^2$

Particular points

For $x = 0$; $v = \pm X_m\omega_0 = \pm V_m$ then $ME = \frac{1}{2}mV_m^2 = \frac{1}{2}mX_m^2\omega_0^2 = KE_{max}$

For $x = \pm X_m$; $v = 0$ then $ME = \frac{1}{2}KX_m^2 = EPE_{max}$

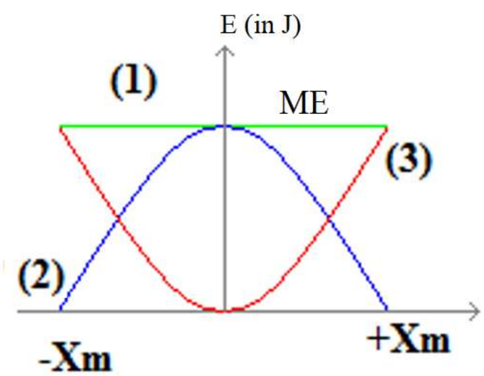
For $x = x_0$; $v = v_0$ then $ME = \frac{1}{2}mv_0^2 + \frac{1}{2}Kx_0^2$

Conservation of *ME*: $\frac{1}{2}mv_0^2 + \frac{1}{2}Kx_0^2 = \frac{1}{2}KX_m^2 = \frac{1}{2}mV_m^2 = \frac{1}{2}mX_m^2\omega_0^2$

VI-Expressions of energies *ME*, *KE* and *EPE* in term of *x*

$ME = const = \frac{1}{2}KX_m^2 = \frac{1}{2}mX_m^2\omega_0^2 = \frac{1}{2}mv_0^2 + \frac{1}{2}Kx_0^2$; $EPE = \frac{1}{2}Kx^2$; $KE = ME - EPE = \frac{1}{2}KX_m^2 - \frac{1}{2}Kx^2$

VII-Variation of energies *ME*, *KE* and *EPE* as function of *x*



$ME = KE + PE$ Graph 1

$KE = \frac{1}{2}mv^2$ Graph 2

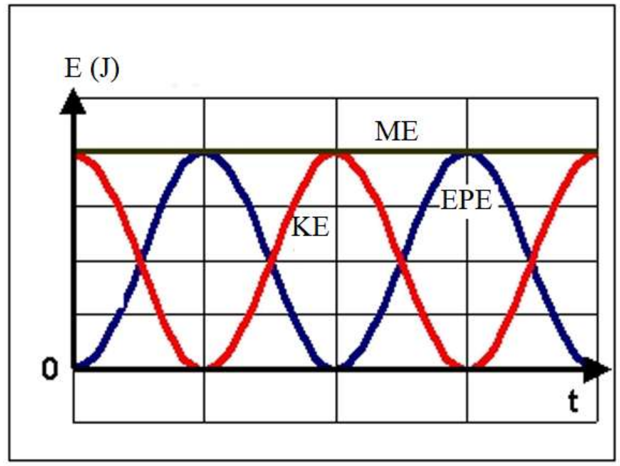
$PE = \frac{1}{2}kx^2$ Graph 3

VIII-Expressions of energies *ME*, *KE* and *EPE* in term of *t*

- $EPE = \frac{1}{2}Kx^2 = \frac{1}{2}KX_m^2 \sin^2(\omega_0 t + \varphi)$
- $KE = \frac{1}{2}mv^2 = \frac{1}{2}mX_m^2\omega_0^2 \cos^2(\omega_0 t + \varphi)$
- $ME = KE + EPE = \frac{1}{2}mX_m^2\omega_0^2 \cos^2(\omega_0 t + \varphi) + \frac{1}{2}KX_m^2 \sin^2(\omega_0 t + \varphi)$

$$ME = \frac{1}{2}KX_m^2 \cos^2(\omega_0 t + \varphi) + \frac{1}{2}KX_m^2 \sin^2(\omega_0 t + \varphi) = \frac{1}{2}KX_m^2 = constant$$

IX-Variation of energies *ME*, *KE* and *EPE* as function of *t*



If $\varphi = 0$

$$EPE = \frac{1}{2}KX_m^2\,sin^2(\,\omega_0 t)$$

$$KE = \frac{1}{2}m\omega_0^2X_m^2\,cos^2(\,\omega_0 t) = \frac{1}{2}KX_m^2\,cos^2(\,\omega_0 t)$$

Note: $Energies\ period = Proper\ period \div 2\ \ or\ T_E = \frac{T_0}{2}$

Application 1

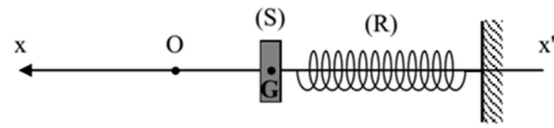
1) Indicate the nature of motion

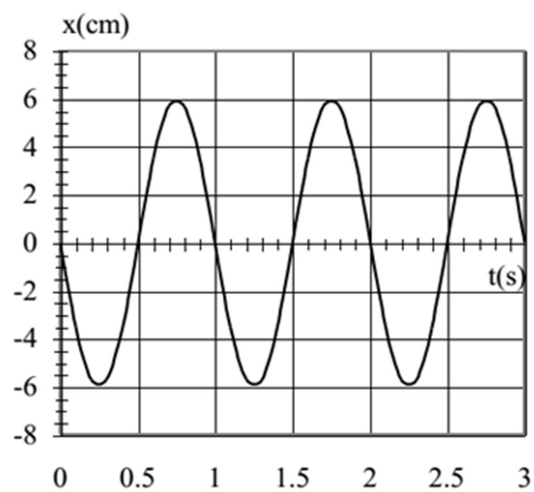
2) Indicate the type (mode) and the regime of the oscillations.

3) Write the expression of *ME* in term of *x* and *v*

`4) Establish the differential equation in *x* that governs motion.

5) The solution of the differential equation is of the form $x(t) = X_m\,sin(\,\omega_0 t + \varphi)$ where X_m ; ω_0 and φ are constant. Determine the expression of ω_0 in term of K and m.





6) Referring to the graph, determine the values of X_m, ω_0 and φ then write the time equation of motion.

7) Calculate the stiffness of the spring knowing that the mass of (S) is m = 160g.

8) Write the velocity time equation. Deduce the value of V_{max}.

9) At what moment does the mobile pass through the equilibrium position *O* for the first time?

10) At which moment t_1 does the speed of the mobile becomes nil for the first time?

11) Determine, by 2 methods, the speed of the mobile when it passes through the abscissa x = 3cm in the negative direction.

12) Trace the shape of the curve representing the variations of the velocity versus *t*.